

Circuit Simulation Lab:11

Design of Latches

Introduction

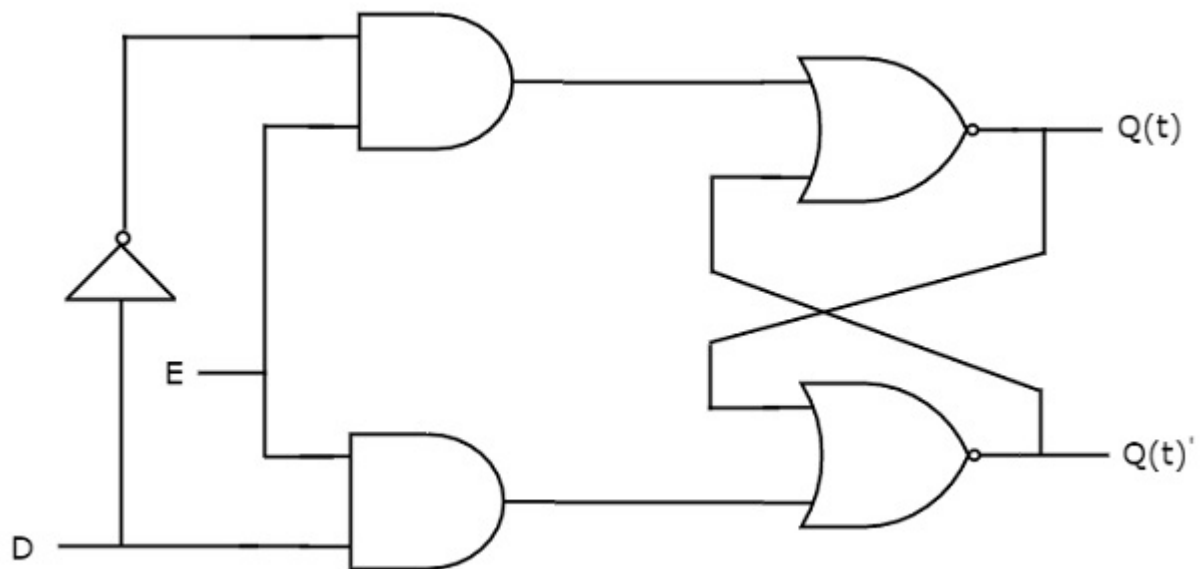
The purpose of this experiment is to introduce the design of simple sequential circuits, in this case Latches. Latch is an electronic device that can be used to store one bit of information. The D latch is used to capture, or 'latch' the logic level which is present on the Data line when the clock input is high. If the data on the D line changes state while the clock pulse is high, then the output, Q, follows the input, D. When the CLK input falls to logic 0, the last state of the D input is trapped and held in the latch.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram

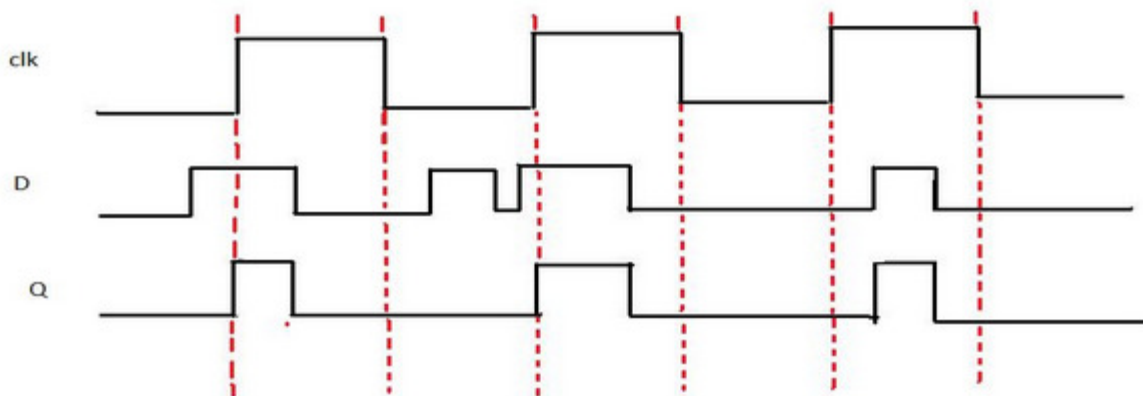


D Latch

D	Q_{t+1}
0	0
1	1

Truth Table

Expected Timing Diagram



Here E is names as clk in the diagram

Describe the above in Verilog HDL and capture the Waveforms

- 1) *Dataflow modeling*
- 2) *Behavior modeling*
- 3) *Structural modeling (Use the gates for designing the Latch)*